

Sujith S

QA Technician | Manual & Automated Test · Test Planning · Grey-Box Test · API Test · Distributed Systems
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Summary

Technical quality assurance practitioner pursuing Int.MSc Computational Statistics and Data Analytics at VIT University (2022-2027), with hands-on experience designing and executing manual and automated test plans for distributed backend services, REST APIs, and consumer-facing web applications. Skilled in grey-box testing – understanding software internals, debugging failures using application logs, and writing automated test scripts in Python – across scalable, cloud-deployed systems on AWS. Experienced coordinating with cross-functional development and product teams to translate technical requirements into detailed test cases and end-to-end regression suites. Strong background in statistical analysis and ML systems provides depth for validating complex, data-driven payment and e-commerce features. Motivated to drive quality-centric development methodologies and deliver defect-free customer experiences at scale.

Education

Int.MSc Computational Statistics and Data Analytics

VIT University | 2022-2027

Coursework: Machine Learning, Deep Learning, Statistical Modeling, NLP, Data Structures & Algorithms, Distributed Systems, Database Systems

CS50 Programs

Harvard CS50 (Online)

Completed [CS50x](#), [CS50P](#), [CS50AI](#), [CS50SQL](#), and [CS50R](#) covering algorithms, AI, Python, and full-stack systems.

Skills

Quality Assurance: Manual Testing, Test Planning, Detailed Test Case Design, Grey-Box Testing, Regression Testing, Functional Testing, End-to-End Testing, UI Automation, Defect Tracking, Root Cause Analysis, Quality-Centric Development

Testing Tools & Automation: Postman (REST API Testing), Python (Test Scripting & Automation), SQL (Data Validation), Bash/Shell, Log Analysis, MLflow, Docker, Linux/Unix

Architecture & Systems: Distributed Systems, Scalable Backend Services, REST APIs, Microservices, PostgreSQL, Redis, Apache Kafka, AWS (Lambda, S3, SageMaker, EKS)

Collaboration & Process: Cross-Functional Stakeholder Coordination, Requirements Analysis, Test Strategy, SLA Definition, CI/CD Pipelines, Agile Methodologies

Languages: Python, SQL, JavaScript, TypeScript, Java, R, Bash/Shell

Work Experience

Founder

[Shoot the Breeze](#) |

2023 - Present

- Designed and executed detailed manual and automated test plans for consumer-facing web applications and REST APIs on AWS; performed grey-box testing by analyzing application logs, debugging failures at the service level, and writing Python test automation scripts to accelerate regression cycles.
- Coordinated with cross-functional stakeholders to understand technical feature implementations and translate requirements into comprehensive end-to-end test suites; identified, documented, and tracked defects through resolution to maintain a high software quality bar before every production release.
- Drove quality-centric development practices across distributed backend services; defined test strategies spanning multiple service boundaries, validated scalable system behavior under realistic usage models, and implemented CI/CD-integrated test automation to catch defects at the earliest stage.

App/Web Developer

[CropSky](#) | VIT-TBI Research & Innovation Hub |

2024 - Present

- Executed functional, regression, and integration testing of ML-powered backend APIs and front-end application features at a university-incubated startup; worked with Software Development Engineers to understand feature implementations and verify that fixes satisfied documented acceptance criteria.
- Developed test cases approximating real-world user scenarios for consumer-facing features; validated large-scale structured data outputs using SQL, debugged failures using service logs, and maintained detailed test result documentation to support production quality targets.

Projects

Demand Forecasting System

[GitHub](#)

Python, FastAPI, PostgreSQL, Redis, MLflow, Docker, AWS, Prophet, XGBoost

- Designed end-to-end test plans for REST inference APIs and forecasting service outputs spanning multiple distributed components; performed grey-box validation by inspecting MLflow logs and service traces, wrote Python automation scripts for regression testing, and defined SLAs for staging-to-production test gates.

Multi-Agent Business Automation System

[GitHub](#)

Python, CrewAI, LangGraph, LangChain, FastAPI, PostgreSQL, Redis, Docker, AWS, Prometheus

- Designed and executed grey-box test suites for multi-step distributed agentic workflows across services owned by independent components; debugged task routing failures using Prometheus metrics and application logs, automated regression tests in Python, and coordinated with engineers to validate defect fixes end-to-end.

Recommendation Engine

[GitHub](#)

Python, PyTorch, FAISS, FastAPI, PostgreSQL, Redis, Docker

- Developed detailed test plans and automated Python test scripts for personalized recommendation APIs; validated functional correctness and latency against real-world usage models, debugged failures at the inference layer using service logs, and drove iterative quality improvements before each production release.

Research & Publications

Stock Market Analysis and Prediction Using Machine Learning

2025

[El Ruha International Conference on Scientific Research](#), pp. 730-740. Designed and executed a rigorous evaluation framework testing 21 ML models across LSTM, GRU, Transformer, ARIMA, XGBoost, LightGBM, and RL agents (DQN, PPO, A2C); validated outputs against performance benchmarks and documented findings in a peer-reviewed publication. [\[GitHub\]](#)

Certifications & Achievements

- [Google Generative AI Leader](#)
- [Google Cloud Computing Foundations](#)
- [FreeCodeCamp Legacy Full Stack](#)
- [OCI 2025 Certified Developer Professional](#)
- AWS [Solutions Architect](#) and [Developer](#)
- [Software Engineer Certification - HackerRank](#)